Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Martindale, Mark Q Ph.D.		
Project Title:	The Evolution of Gravity Reception in the Metazoa		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jacobs, Dave (U.C.L.A.)		
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	We will investigate the evolutionary origins of gravity detection in multicellular organisms. The most basal metazoans capable of gravity reception are members of the Cnidaria and Ctenophora. The apical organ of ctenophores and the rhopalia of pelagic medusoid cnidarians are composed of modified ciliary structures and share many structural similarities to the gravity detectors of higher animals, such as vertebrate vestibular hair cells. In order to determine the possible evolutionary links between gravity amplifiers in the history of life on earth, we intend to study representatives from these two pivotal phyla. We have isolated several genes (e.g. forkhead and hairy/enhancer of split orthologs in ctenophores and brain/POU genes in both ctenophores and cnidarians) that are expressed in the developing sense organs and will continue to isolate candidate genes by PCR and library screening involved in gravity detection. Specific targets are Otx, atonal, bHLH		

Task Description:	genes, Pax, and notch orthologs, all of which are involved in sensory cell development in other systems. The spatial and temporal expression of these genes will be determined, and address the function investigated by RNAi and morpholino gene inactivation techniques. Reagents and cell specific markers will be developed to allow future physiological studies of the gravity detection apparatus. Because many of these genes are also expressed in other sensory structures our experiments will also shed light on to the evolutionary relationship between gravity detection and other sensory modalities such as ?hearing? and the ability to detect light. Our work will not only use rigorous phylogenetic reconstruction techniques to help understand the evolution of gravity detection in the Metazoa but will also determine the phylogenetic relationships of these basal metazoan groups to bilaterian forms. The complimentary backgrounds of the two Co-P.I.s, the experience with the development and anatomy of the study organisms, in addition to the preliminary data from both labs, insures that rapid progress on these topics will be made.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is the first systematic investigation of the evolution of sensory strucutres involved in detecting position in the earth's gravitational field. Our experimental design is intended to determine ancestral and derived aspects of this process and gain a better idea of the developmental and molecular potentialities of ancestral genomic programs.
Task Progress:	We have made progress on both fronts. MQM's lab established homology of the oralaboral axis of cnidarians and ctenophores to the bilaterian anterior-posterior axis. Using Hox gene expression in the anthozoan Nematostella vectensis, we published a paper in Science, that argued that the oral pole (mouth) of the oral-aboral axis (the major longitudinal body axis of ctenophores and cnidarians) is homologous to the anterior end and the opposite (aboral pole) to the posterior end of bilaterian body axes. Thus, the apical organ in ctenophores, situated on the aboral pole, is on the OPPOSITE end of the animal than the 'brain' is located in higher animals. This suggests that the apical organ of ctenophores is NOT homologous with bilaterian neural structures, as has been assumed by virtually every evolutionary scenario on the evolution of the bilaterian body. A similar conclusion was made in Nematostella using a marker for olfactory neurons in bilaterians. COE gene expression in Nematostella was found in the apical tuft of sensory cilia located on the aboral pole (side opposite the mouth), showing that in both cnidarians and ctenophores, neural structures could form in 'non anterior' locations of the adult body plan. These results, surprising as they may be, are corroborated by in situ expression analysis of candidate neural genes in ctenophores. Although 'negative' data, of 5 newly isolated candidate genes three sine oculis orthologs, and two zinc-finger containing transcription factors, only one of the zinc-finger genes was expressed in progenitors of the apical organ (Pang and Martindale, in preparation). This is consistent with two additional neural genes (Ct-vax and Ct-T-brain) and a previousl published neural marker ('brain factor-3' (Yamada and Martindale, 2002) that were also not expressed in the apical organ. We have isolated several new genes (2 nanos orthologs, vasa, 2 barH homeodomain orthologs, mef2, aristaless, and otx) that we are continuing to characterize to see if the pattern continues. Despite the fact that the apical organ does not seem to be homologous to the bilaterian brain, it provides us with tremendous opportunities to study the evolution of gene expression and the co-option of existing genes for novel functions (this is, after all, the driving force for the molecular mechanisms generating evolutionary novelty). One particularly interesting example recently published in Nature (Kusserow et al 2005) was the study of the Wnt family of signaling molecules in Nematostella. We showed that this 'simple' animal has essentially the full complement of Wnt genes as vertebrates, and were able to demonstrate a novel (and potentially ancestral) colinear pattern of family members in both the ectoderm as well as the endoderm. The implications of these data are that there is a disconnect between the genomic complexity and the morphological complexity of animal body plans, raises questions about the nature of individual 'cell types', and indicates that the genomes of many animals (such as the 'model' systems of fruit flies and soil nematodes) have highly modified genomes that have apparently lost many gene family members. Other advances related to this project include the completion of the sequencing of the entire genome of the anthozoan Nematostella vectensis by the JGI (Dept. of Energy). Assembly is beginning in January of 2005 and should be completed within several months. A ctenophore, likely to be Mnemiopsis leidyi, although other spedciea re under consideration) has been proposed for total genome sequencing in early 2006 (Dan Rokhsar, JGI). My laboratory has already shipped high quality genomic DNA and RNA from Mnemiopsis to JGI for preliminary work. We have already begun to use the genome data to isolate important genes from Nematostella and we are anxious to receive molecular data from a ctenophore as soon as it is available. In 2004 the Jacobs lab continued to make progress as regards the objectives of the grant. We published a paper on the recovery of the sense organ regulatory gene sine oculis in a range of basal metazoan taxa including jellyfish and sponges -including observation of expression of the gene in jellyfish (Bebenek et al. 2004). We continue to recover genes involved in the regulatory cascade controlling sense organ developmental. These genes have not been previously observed in Basal Metazoa. Nagayasu Nakanishi a student in the Jacobs lab recovered the eyes absent gene and has conducted in situ hybridization work with this gene in jellyfish. An additional gene Optix has now been recovered from a number of basal Metazoa including Jellyfish and sponges. Nagayasu have also been using more general neural and muscle specific fluorescently labelled probes to better characterize the sensory and muscular structures in the planula , polyp and ephyra life history stages. These probes are examined using confocal microscopy to yield very detailed reconstructions of the muscular and neural organization in the sensory structures found in these distinct life history stages. With these detailed morphologic studies we can much better characterize the precise locate the cells in which gene expression is occurring. We have also recovered the sense organ specific neurogenic gene atonal and the head regulator orthodenticle.
Bibliography Type:	Description: (Last Updated: 12/26/2007)
Articles in Peer-reviewed Journals	Henry, J. Q. and Martindale, M.Q. "Inductive Timing and Equivalence Groups in the Ctenophore Mnemiopsis leidyi" Evol. and Develop. 6, 17-24 , Jan-2004
Articles in Peer-reviewed Journals	Pang, K., Matus, D.Q., and Martindale, M.Q. "The ancestral role of COE helix loop helix genes was in chemoreception: Evidence from the development of the sea anemone, Nematostella vectensis (phylum, Cnidaria; Class Anthozoa)." Develop. Genes and Evol. 214, 134-138 , Jan-2004
Articles in Peer-reviewed Journals	Finnerty, J.R., Pang, K., Burton, P., Paulson, D., and Martindale, M.Q. "Deep Origins for Bilateral Symmetry: Hox and Dpp Expression in a sea anemone" Science. 304, 1335-1337 , Jan-2004

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